# PyLumerical Cheat Sheet

**Version: main**

## Install with virtual environment

```
python -m venv .venv
.venv\Scripts\Activate.ps1 # Windows PowerShell.
    For Linux, run source .venv/bin/activate
python -m pip install -U pip
python -m pip install ansys-lumerical-core
```

## Starting a session

Start a Lumerical FDTD session

```
import ansys.lumerical.core as lumapi
with lumapi.FDTD() as fdtd:
  print(f"This is Lumerical FDTD {fdtd.version()}")
  input("Press any key to close session")
```

Start Lumerical Multiphysics, add CHARGE and HEAT solvers

```
import ansys.lumerical.core as lumapi
with lumapi.DEVICE() as device:
  device.addchargesolver()
  device.addheatsolver()
```

Start a Lumerical INTERCONNECT session, and print library

```
import ansys.lumerical.core as lumapi
with lumapi.INTERCONNECT() as intc:
  print(intc.library())
```

The `hide` flag hides the GUI during execution

```
import ansys.lumerical.core as lumapi
with lumapi.FDTD(hide=True) as fdtd:
  print(f"This is Lumerical FDTD {fdtd.version()}")
```

Run existing scripts on an existing Lumerical FDTD project

```
with lumapi.FDTD(hide=True,
    script="SetupScript.lsf",
    project="MyPyLumProject.fsp") as fdtd:
  # Use function from SetupScript.lsf
  fdtd.customsetupfunction()
  # Do analysis after running simulation
  analysisScript =
      open("AnalysisScript.lsf","r").read()
  fdtd.eval(analysisScript)
```

## Set up simulation objects

Using keyword arguments

```
with lumapi.FDTD() as fdtd:
  fdtd.addfdtd(dimension="2D", x=0.0e-9, y=0.0e-9,
      x_span=3.0e-6, y_span=1.0e-6)
```

Using set commands

```
with lumapi.FDTD() as fdtd:
  fdtd.addrect({"name": "MyRect"})
  fdtd.setnamed("MyRect","x", 0)
  fdtd.setnamed("MyRect","x span", 1e-6)
```

Using `OrderedDict` to preserve property assignment order

```
from collections import OrderedDict
with lumapi.FDTD() as fdtd:
  props = OrderedDict([("name",
      "power"),("override global monitor
      settings", True),("x", 0.),("y", 0.4e-6),
      ("monitor type", "linear x"),("frequency
      points", 10.0)])
  fdtd.adddftmonitor(properties = props)
```

## Running simulations

Set up resources

```
with lumapi.FDTD(hide=True) as fdtd:
  # Add CPU/GPU resources
  cpuRes = fdtd.addresource("FDTD")
  fdtd.setresource("FDTD", cpuRes, "device type",
      "CPU")
  fdtd.setresource("FDTD", cpuRes, "name", "PyLum
      CPU")
  gpuRes = fdtd.addresource("FDTD")
  fdtd.setresource("FDTD", gpuRes, "device type",
      "GPU")
  fdtd.setresource("FDTD", gpuRes, "name", "PyLum
      GPU")
```

Run simulation locally

```
with lumapi.FDTD() as fdtd:
  # ... Create your project
  fdtd.save("MyPyLumericalProject.fsp")
  fdtd.run("FDTD", "GPU", "PyLum GPU") # Run GPU
      solver on the PyLum GPU resource
```

Run FDTD solver on Ansys Cloud Burst Compute™

```
with lumapi.FDTD() as fdtd:
  burstSettings={"account": "user@company.com",
      "download": True, "Name": "MyPyLumBurstJob"}
  fdtd.run("FDTD", "GPU", "burst", burstSettings)
```

Run a MODE simulation

```
with lumapi.MODE() as mode:
  mode.addfde()
  mode.setanalysis("wavelength", 1.55e-6)
  mode.setanalysis("search", "near n")
  mode.save("MyPyLumMODEProject.lms")
  mode.findmodes()
```

Run a sweep

```
with lumapi.FDTD(hide=True) as fdtd:
  fdtd.addsweep(0) # 0 - Sweep, 1 - Optimization,
      2 - Monte Carlo, 3 - S-parmaeter Matrix, 4
      - Corner
  fdtd.setsweep("sweep","name","childSweep")
  fdtd.setsweep("childSweep", "type", "Ranges")
  fdtd.setsweep("childSweep", "number of points",
      10)
  # Add parameters to existing object named film
  sweepParams={"Name":"thickness", "Parameter":
      "::model::film::z max", "Type": "Length",
      "Start":0.05e-6, "Stop": 0.15e-6}
  fdtd.addsweepparameter("childSweep",sweepParams)
  # insertsweep adds as a parent for sweeps, child
      for other types
  fdtd.insertsweep("childSweep")
  fdtd.setsweep("sweep","name","parentSweep")
  # Add result to existing monitor named monitor
  sweepResult1={"Name": "T", "Result":
      "::model::monitor::T"}
  fdtd.addsweepresult("parentSweep", sweepResult1)
  fdtd.runsweep("parentSweep")
```

## Visualize results

Visualize monitor data using matplotlib Python library

```
import numpy as np
import matplotlib.pyplot as plt
with lumapi.FDTD(hide=True) as fdtd:
  # Retrieve result from Lumerical dataset
  eFieldRes = fdtd.getresult("monitor", "E")
  x, y = eFieldRes['x'] , eFieldRes['y']  #
      Convert to um
  Ex, Ey, Ez = eFieldRes['E'][:,:,0,0,0],
      eFieldRes['E'][:,:,0,0,1],
      eFieldRes['E'][:,:,0,0,2]
  # fdtd.getelectric() also gets amplitude
  eFieldAmplitude = Ex ** 2 + Ey ** 2 + Ez ** 2
  X, Y = np.meshgrid(x, y)  # Create meshgrid
  plt.contourf(X, Y, eFieldAmplitude)
```